

BDT for HL-LHC Muon Track Isolation

Sergei Uzunian, Jahred Adelman, Kevin Sedlaczek



Northern Illinois University



Introduction

- ▶ Undergraduate at Northern Illinois University
- ▶ Prompt vs Non-Prompt Muon Track Isolation
- ▶ Simulated ATLAS HL-LHC data with $\langle\mu\rangle = 200$
- ▶ Results from XGBoost-Based BDT Classifier Model
- ▶ Comparing with Cut-Based Isolation
- ▶ Comparing with Neural Network Approach

Motivation

- ▶ LHC today runs at average pileup $\langle\mu\rangle \approx 50$
- ▶ HL-LHC: $\langle\mu\rangle \approx 200$ collisions per bunch crossing
- ▶ Many more overlapping tracks per event
- ▶ Harder to separate muons (for example in W , Z decays) from non-prompt muons in b/c -hadron jets
- ▶ Motivates more sophisticated muon isolation and discrimination tools
- ▶ BDT offers relatively simple improved approach

Sample Information

- ▶ Signal — single prompt muon event with pileup $\langle\mu\rangle = 200$
 - ▶ Muon added directly without simulated decay event
 - ▶ 9,291 events used, $\sim 1,900$ tracks/event
 - ▶ Each signal event contains one prompt muon
- ▶ Background — HH(4b) events
 - ▶ Non-prompt muons from b -hadron decays inside jets
 - ▶ 9,999 events used, $\sim 1,970$ tracks/event
- ▶ Each event stores list of reconstructed inner-detector tracks
- ▶ Per-track kinematics
 - ▶ p_T, η, ϕ
- ▶ Per-track impact parameters
 - ▶ $z_0 \sin \theta$ (longitudinal), d_0 (transverse)
- ▶ `isMuon` flag tags which tracks are reconstructed muons

BDT Input Features

- ▶ Basic Features of Muon
 - ▶ p_T — transverse momentum
 - ▶ η — pseudorapidity
 - ▶ $|z_0 \sin \theta|$ — longitudinal impact-parameter magnitude
 - ▶ d_0 — transverse impact-parameter
- ▶ Neighboring Track Features
 - ▶ From ΔR and $|\Delta z_0 \sin \theta|$ based cone of ($K = 10$) tracks
 - ▶ Rankings: ΔR -ascending and p_T -descending
 - ▶ Each slot carries: d_0 , p_T , η , ΔR -to-candidate, $|z_0 \sin \theta|$
- ▶ Additional Optional Features
 - ▶ Isolation scalar (cone p_T sum / muon p_T)
 - ▶ Used in showcased figures
 - ▶ Zeta ordering: $\zeta = \sqrt{(20 \Delta R)^2 + (\Delta z_0 \sin \theta)^2}$

BDT Training Methodology

- ▶ Drop tracks with $p_T < 1$ GeV
- ▶ Two kinds of reweighting (one per-muon training weight)
 - ▶ Kinematic — GBReweigher weights background so muon (p_T, η) matches signal
 - ▶ Prevents separating on sample kinematics instead of isolation; fit on training set only
 - ▶ Class imbalance — each signal muon upweighted by $n_{\text{bkg}}/n_{\text{sig}} \approx 62$ ($\sim 465\text{k}$ bkg vs $\sim 7.5\text{k}$ sig training muons)
 - ▶ Final weights: signal carries imbalance factor, background carries kinematic weight
- ▶ Neighboring Tracks Padding
 - ▶ XGBoost library handles missing values natively
- ▶ 80/20% split for training/ holdout (by event)
 - ▶ Train (80%) — fits the trees
 - ▶ Holdout (20%) — unseen muons
 - ▶ Holdout used for performance metric

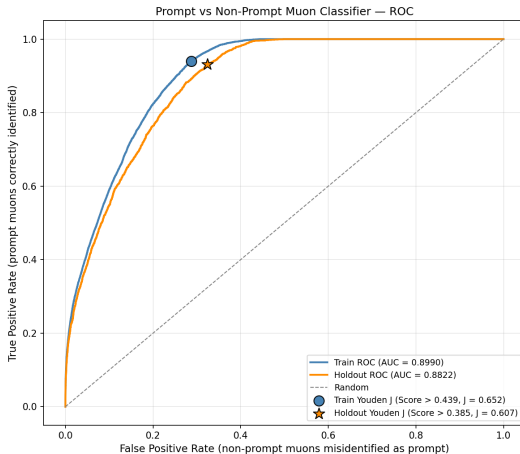
Evaluation — ROC and AUC

- ▶ ROC curve: True Positive Rate vs False Positive Rate as the score threshold is swept
 - ▶ $\text{TPR} = \text{prompt muons correctly identified} / \text{all prompt muons}$
 - ▶ $\text{FPR} = \text{non-prompt muons misidentified as prompt} / \text{all non-prompt muons}$
 - ▶ A better classifier bows toward the top-left corner
- ▶ AUC = area under the ROC curve (0.5 = random, 1.0 = perfect)
- ▶ Training AUC — computed on the 80% muons seen by model
- ▶ Holdout AUC — computed on the 20% unseen muons
- ▶ Training/Holdout AUC gap used as overfitting benchmark

Youden's J Statistic

- ▶ A distinguished point chosen on the ROC curve
- ▶ Youden's $J = \text{TPR} - \text{FPR}$
- ▶ Ranges from 0 (random) to 1 (perfect separation)
- ▶ Geometrically the farthest point above the random diagonal
- ▶ Chosen threshold for BDT classification score maximizes J
 - ▶ Weights correct identification of prompt muons and rejection of non-prompt muons equally
- ▶ Provides another meaningful score alongside the AUC

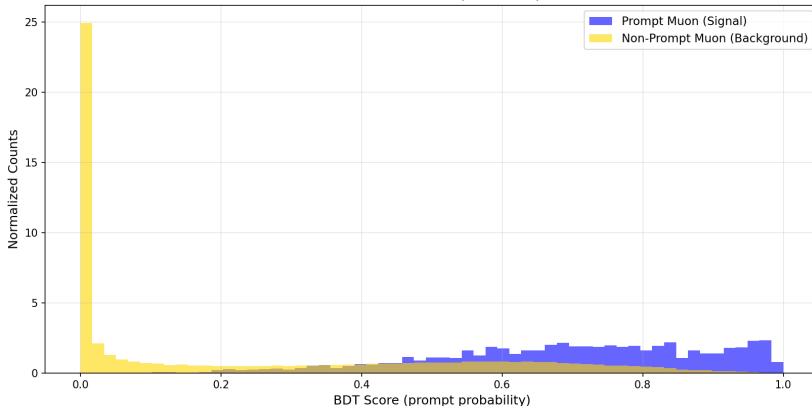
BDT Performance — ROC



```
RUN SETTINGS (muonBDT.py v1.0)
Samples      : signal=run_InvPTPU200 (prompt), bkg=run_bjet (non-prompt); label=provenance
Candidates  : muons [isMuon], pT >= 1.0 GeV; bkg all muons/event, signal 1/event
Features     : muon pT, eta, |z0sinθ|, d0 (signed), isolation; neighbors carry d0, pT, Δeta, ΔR, Δz0sinθ
Isolation    : ΣpT(cone tracks)/pT(muon); cone = dR<0.5 and |z0sinθ track-z0sinθ muon|<10.0mm, pT>=1 GeV, muon excluded
Neighbors    : <=K=10 same-event tracks, dR<0.5 and |Δz0sinθ|<10.0mm (signed vertex sep); same cone train/holdout; dR_asc & pT_desc; NaN-padded
Reweighting : GBReweighter (pT,eta) bkg->signal, train only
Imbalance   : scale_pos_weight=61.62 on signal
Events      : signal 9,291 events / 9,291 muons, background 9,999 events / 570,694 muons
Train/holdout : 465,378 / 114,607 muons (80/20 split by event)
Hyperparams  : trees=200, depth=3, lr=0.1, subsample=0.8, colsample=0.8, min_child_weight=1.0, reg_lambda=1.0
```

BDT Output Score

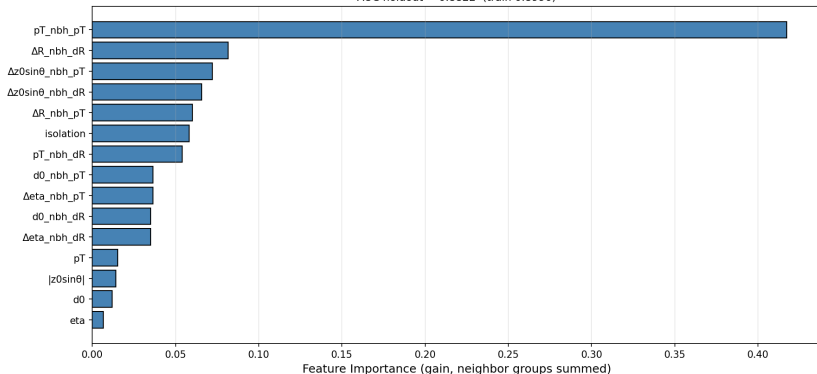
Holdout BDT Score — Prompt vs Non-Prompt Muons
AUC holdout = 0.8822 (train 0.8990)



```
RUN SETTINGS (muonBDT.py v1.0)
Samples      : signal=run_InvTPU200 (prompt), bkg=run_bjet (non-prompt); label=provenance
Candidates   : muons (isMuon), pT >= 1.0 GeV; bkg all muons/event, signal 1/event
Features     : muon pT, eta, |z0sinθ|, d0 (signed), isolation; neighbors carry d0, pt, Δeta, ΔR, Δz0sinθ
Isolation    : 1pT(cone tracks)/pT(muon); cone = dR<0.5 and |z0sinθ track-z0sinθ muon|<10.0mm, pT>=1 GeV, muon excluded
Neighbors    : <=K=10 same-event tracks, dR<0.5 and |Δz0sinθ|<10.0mm (signed vertex sep); same cone train/holdout; dR-asc & pT-desc; NaN-padded
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```

BDT Feature Importances

Feature Importances — Prompt vs Non-Prompt Muon BDT
AUC holdout = 0.8822 (train 0.8990)



neighbor feature key (nbh = up to K=10 tracks in the neighbor cone, same event)

<var> nbh_dR : ranked by ΔR ascending (closest first)

<var> nbh_pT : ranked by pT descending (hardest first)

each nbh bar = sum of gain over all K slots of that (var, ordering)

RUN SETTINGS (muonBDT.py v1.0)

Samples : signal=run InvPU200 (prompt), bkg=run bjet (non-prompt); label=provenance

Candidates : muons (isMuon), pT >= 1.0 GeV; bkg all muons/event, signal 1/event

Features : muon pT, eta, |z0sinθ|, d0 (signed), isolation; neighbors carry d0, pt, Δeta, ΔR, Δz0sinθ

Isolation : IpT(cone tracks)/pT(muon); cone = dR<0.5 and |z0sinθ_track-z0sinθ_muon|<10.0mm, pT>=1 GeV, muon excluded

Neighbors : <K=10 same-event tracks, dR<0.5 and |Δz0sinθ|<10.0mm (signed vertex sep); same cone train/holdout; dR-asc & pT-desc; NaN-padded

Reweightings : GBReweighter (pT,eta) bkg->signal, train only

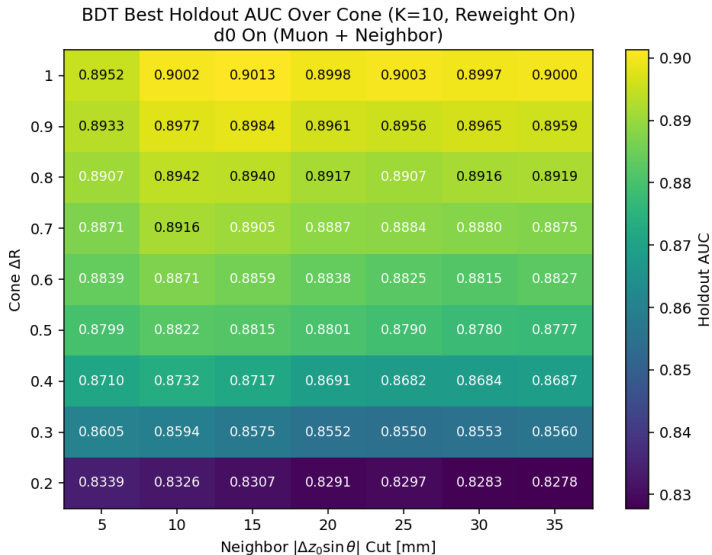
Imbalance : scale_pos_weight=0.62 on signal

Events : signal 9,291 events / 9,291 muons, background 9,999 events / 570,694 muons

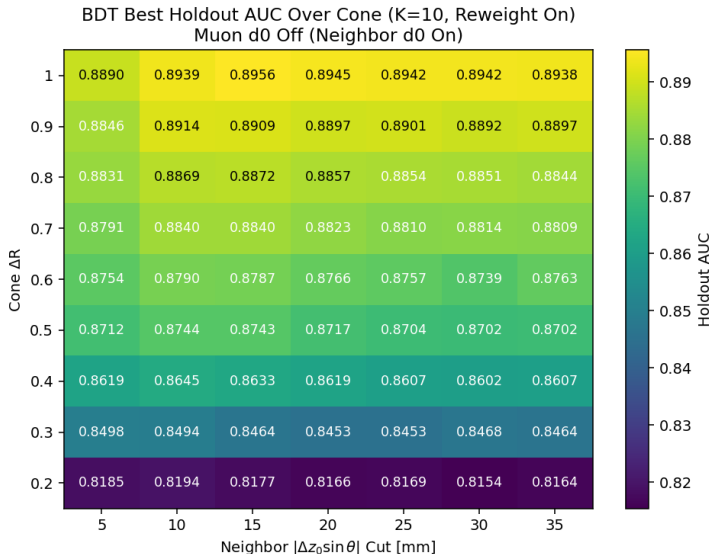
Train/holdout : 465,378 / 114,607 muons (80/20 split by event)

Hyperparams : trees=200, depth=3, lr=0.1, subsample=0.8, colsample=0.8, min_child_weight=1.0, reg_lambda=1.0

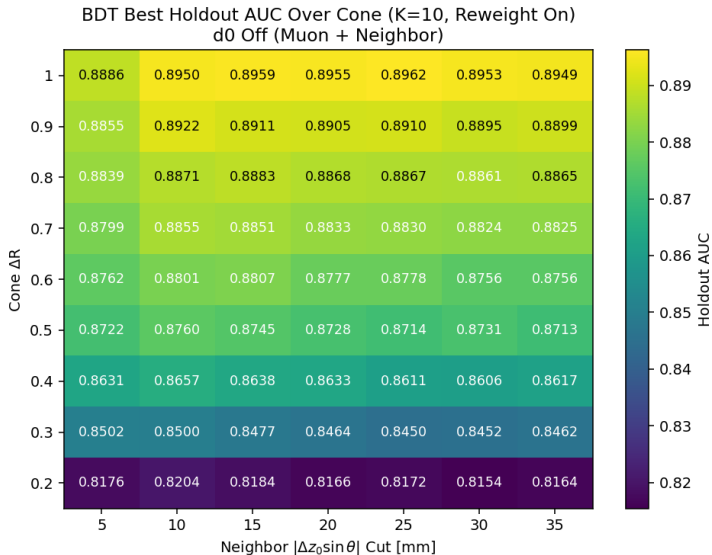
Cone Scan — d_0 On



Cone Scan — Muon d_0 Off



Cone Scan — d_0 Off



Effect of d_0 and Optional Features

- ▶ Muon's own d_0 — small gain
 - ▶ Holdout AUC 0.8822 (on) vs 0.8744 (off), $\Delta \approx +0.008$
- ▶ Neighbor-track d_0 — negligible
 - ▶ 0.8744 (neighbor d_0 on) vs 0.8760 (all d_0 off), $\Delta \approx 0$
- ▶ Best model keeps muon d_0 information
- ▶ Neighbor d_0 information can be dropped
- ▶ Isolation scalar and zeta ordering give negligible AUC gain
 - ▶ Isolation scalar: +0.000; zeta ordering: +0.0004
 - ▶ Already captured by the neighbor-track features

Cut-Based Isolation

- ▶ Given feature importances, most of the discrimination appears to come from the surrounding track activity
- ▶ Motivates comparing against cut-based track isolation
- ▶ Consider relative track isolation $\sum p_T^{\text{cone}}/p_T^\mu$ of muon by including nearby tracks in a ΔR and $|\Delta z_0 \sin \theta|$ based cone
- ▶ Cone ΔR and $|\Delta z_0 \sin \theta|$ window scanned for the best-performing isolation value
- ▶ A proxy for any cut-based isolation tool

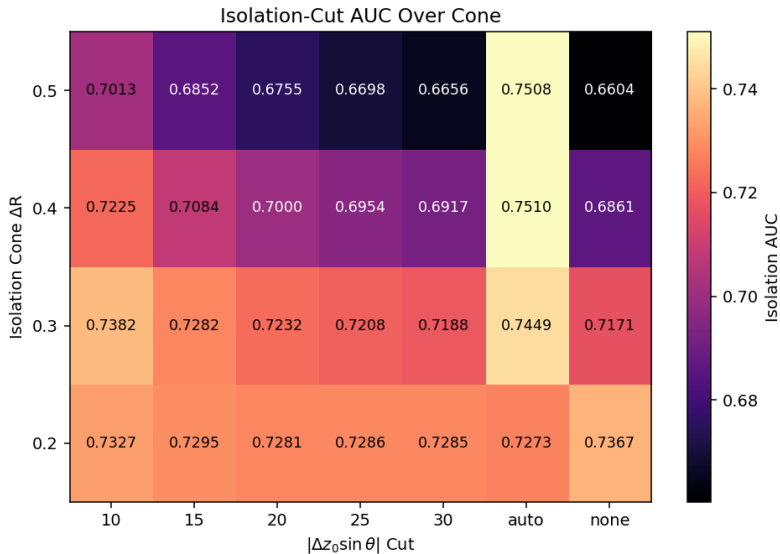
Isolation Window Requirement

p_T [GeV]	$ \eta < 1$	$1 < \eta < 2$	$ \eta > 2$
< 1.3	0.6	0.6	0.5
1.3–1.5	0.6	0.6	0.6
1.5–2.0	1.0	1.0	0.6
2.0–3.0	1.2	1.2	1.0
3.0–4.0	2.0	1.6	1.0
4.0–5.0	3.3	2.5	1.6
5.0–6.0	4.2	3.3	2.5
6.0–7.0	5.4	5.2	4.2
7.0–10.0	6.0	5.2	4.2
10.0–20.0	6.0	5.5	4.2
20.0–30.0	6.0	6.6	5.4
> 30.0	6.0	8.5	8.5

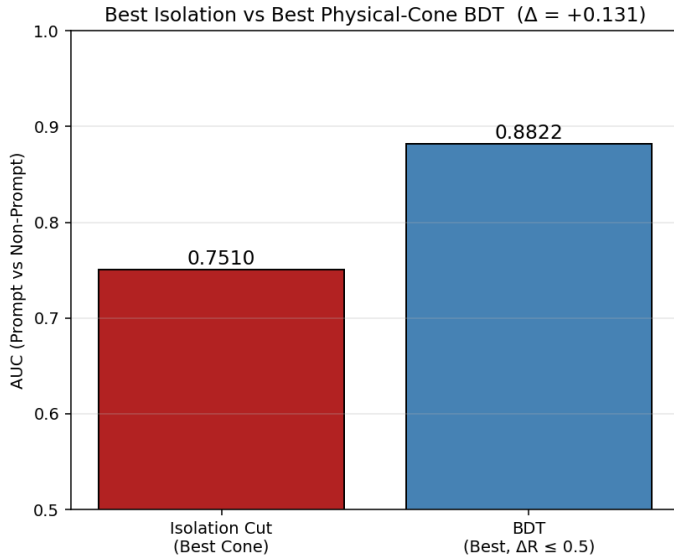
Table C.1: per-track $|\Delta z_0 \sin \theta|$ window for pileup rejection, binned in track p_T and $|\eta|$

- ▶ Lets the physics-motivated window be compared against plain fixed ΔR / $|\Delta z_0 \sin \theta|$ cuts
- ▶ Next slide's “auto” column sets each track's $|\Delta z_0 \sin \theta|$ cut from this table, instead of one fixed value
- ▶ Isolation AUC is best at small cones and with the “auto” window (≈ 0.75)
- ▶ AUC falls as the cone ΔR widens and as the $|\Delta z_0 \sin \theta|$ window loosens

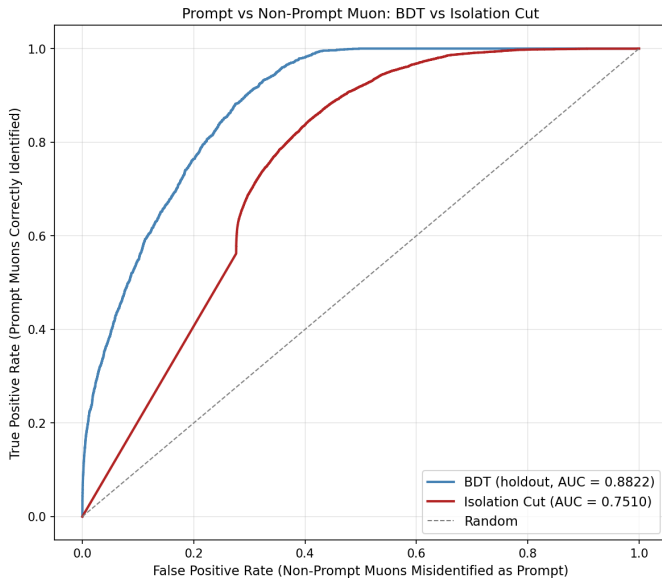
Isolation Cut — Cone Scan



BDT vs Cut-Based Isolation



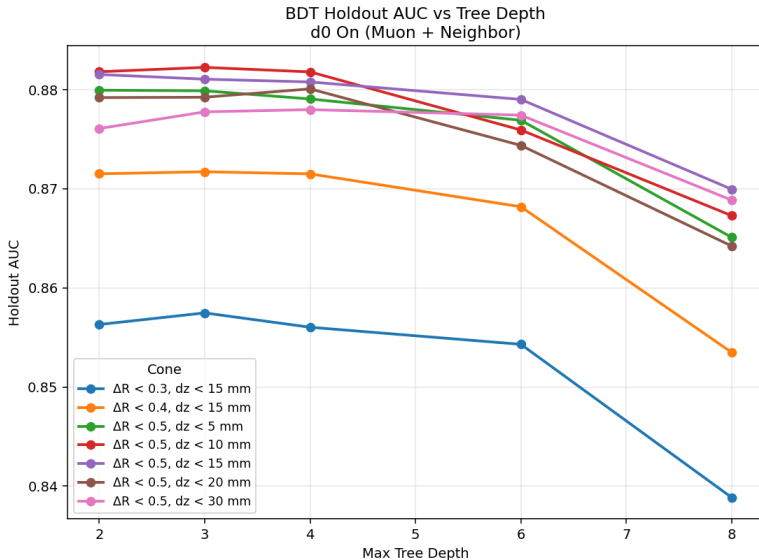
BDT vs Isolation — ROC



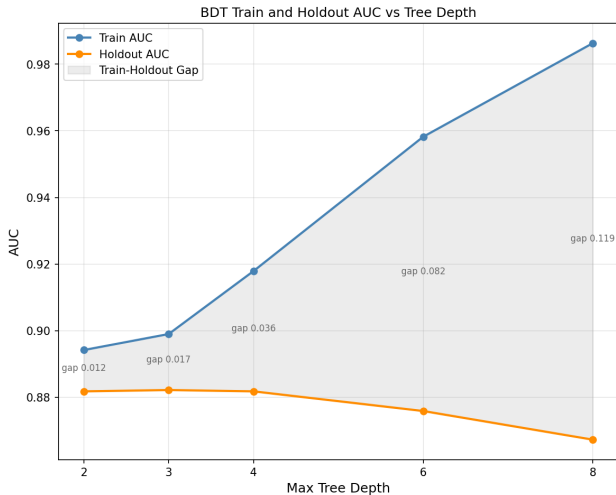
Further BDT Observations

- ▶ AUC decreases with tree depth $d > 3$
- ▶ AUC does not plateau past the physical cone
- ▶ Increased ΔR may capture more physics of neighboring tracks
- ▶ Prima facie, BDT performs about the same or better as (work in progress) neural network approach

BDT Tree-Depth Scan



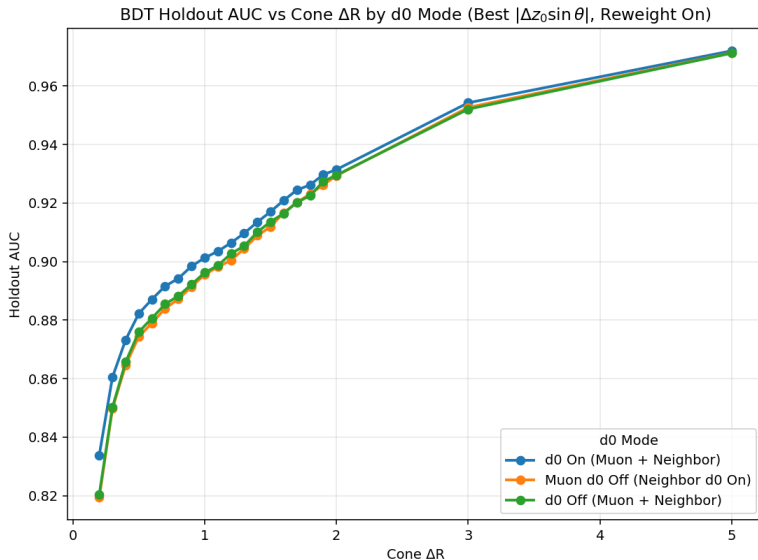
BDT Train-Holdout Gap



RUN SETTINGS

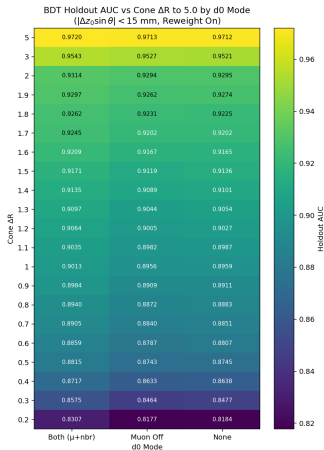
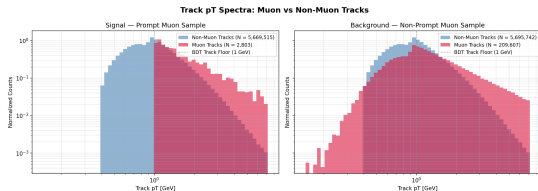
Samples : signal=run_InvPtPU200 (prompt), bkg=run_bjet (non-prompt); label=provenance
Cone : dR<0.5 and $|\Delta z \sin \theta| < 10.0 \text{ mm}$, K=10 neighbors, d0-mode=both, isolation on
Reweighting : GBReweighter (pT,eta) bkg->signal, train only; scale_pos weight on signal
Selection : best holdout AUC per tree depth across dz10 d0on scans (Full statistics)

BDT AUC vs Cone Size

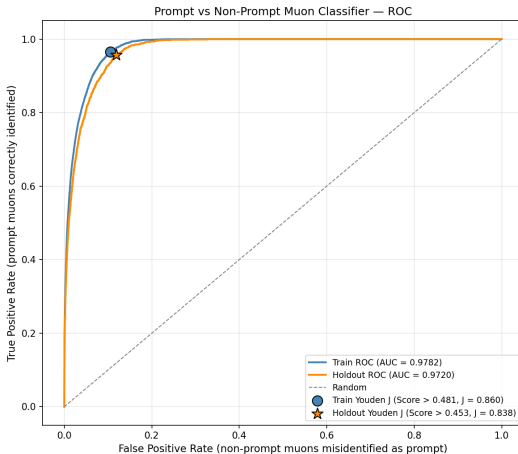


Extending the Cone — No AUC Plateau

- ▶ Extending $\Delta R \rightarrow$ AUC keeps rising, no plateau
- ▶ No noticeable difference in sample p_T distributions



BDT at a Wide Cone

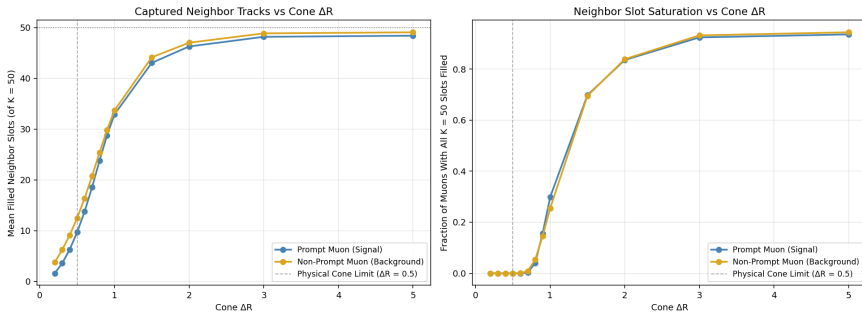


```

RUN SETTINGS (muonBDT.py v1.1)
Samples      : signal=run_InvPtPU280 (prompt), bkg=run_bjet (non-prompt); label=provenance
Candidates   : muons (isMuon), pT >= 1.0 GeV; bkg all muons/event; signal 1/event
Features      : muon pT, eta, |z0sinθ|, d0 (signed), isolation; neighbors carry d0, pt, Δeta, ΔR, Δz0sinθ [d0-mode=both]
Isolation     : ipT(cone tracks)/pT(muon); cone = dR<5.0 and |Δz0sinθ-track-z0sinθ_muon|<15.0mm, pT>=1 GeV, muon excluded
Neighbors     : <=K=10 same-event tracks, dR<5.0 and |Δz0sinθ|<15.0mm (signed vertex sep); same cone train/holdout; dR-asc, pT-desc; NaN-padded
Reweighting   : GBReweighter (pT,eta) bkg->signal, train only
Imbalance     : scale_pos_weight=0.62 on signal
Events        : signal 9,291 events / 9,291 muons, background 9,999 events / 570,694 muons
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```

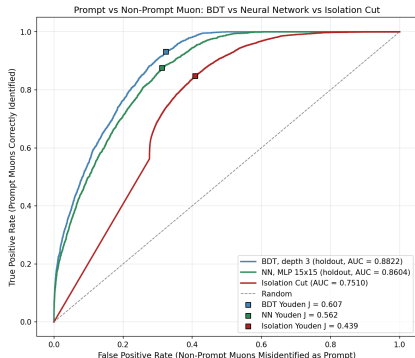
Neighbor Capture vs Cone Size

BDT Neighbor Capture ($K = 50$, $dz < 15$ mm)



BDT vs Neural Network

- ▶ Same inputs and preprocessing as the BDT
 - ▶ Muon + up to $K = 10$ neighbors, GBReweigher, 80/20 split
 - ▶ Features zero-padded and standardized
- ▶ PyTorch MLP:
 - 105 $\rightarrow$ 15 $\rightarrow$ 15 $\rightarrow$ 1
 - ▶ ReLU hidden layers, $\sim 1,850$ parameters
 - ▶ Weighted cross-entropy, Adam, 21 epochs
- ▶ Current best (physical cone): holdout AUC 0.8604
 - ▶ BDT performs 0.8822 at the same cone
 - ▶ NN may still outperform once better optimized



```
RUN SETTINGS
Samples : signal+run_InvPUP200 (prompt), bkg+run_bjet (non-prompt); label=provenance
Cone (MC) : dR=0.5 and [abs(sini)-lEm, K=10 neighbors, dR=both, isolation on
BDT : XGBoost trees=200, depth=3, lr=0.1; NaN-padded neighbor slots
NN : MLP [105->15->15->1, ReLU, Adam lr=0.001, 21 epochs; zero-padded, standardized
Isolation : 2pT(cone)/pT(buon), best cone dR=0.4 with 'auto' [abs(sini)] window
'auto' = per-track window from the (track pT, [eta]) table
Reweighting : GBReweigher (pT,eta) bkg->signal, train only; scale_pos_weight on signal
Evaluation : BDT and NN on the same 20% holdout (split by event); isolation is untrained
Youden J : box-marked points at max(FPR-FNR) on each curve
```

Acknowledgements

- ▶ US-ATLAS SUPER
- ▶ Jahred Adelman
- ▶ Kevin Sedlaczek
- ▶ NIU METIS

Project Repository

- ▶ Code and results for this project are on GitHub:
 - ▶ <https://github.com/sergeiuzunian/muon-isolation>